

CULTIVARS

- Napoli
- Adelaide (early)

INTENSIVE SPACING

- 6 rows (5"-12 cm) per bed spaced every 2" (5 cm) on the row
- Size 1.8-2.0: JANG 3X seeder: X-24, 11 F-13B, brush "0", ≠

FERTILIZATION

- Alfalfa meal (2-0-2) or chicken manure (5-3-2) in a bed where the previous crop was amended with compost.

IRRIGATION

Irrigation Type

- Drip tape (4 per beds) or sprinkler

Irrigation Management

- Root depth: Medium (24"-40")
- Need deep watering until later stages of root development, at which time excess water can cause roots to crack. Cracking is also caused by too great a fluctuation between wet and dry.

IMPLANTATION

Generally, we seed 2 weeks worth of carrots every 1 weeks to ensure a continuous supply of freshly harvested carrots. Every time we seed, we prepare next week's beds in order to force weeds, according to our pre-emergence flame weeding strategy.

Our spacing (6 rows, 2' on the row) means that all the carrots are ready more or less at once. They are then graded in the fields and the bed is cropped out and replanted faster.

EQUIPMENT

- Broadfork
- Rotary harrow
- Jang seeder 3X + X-24 rollers, scale, notepad and seeds
- 30" rake
- Flame weeder

ITINERARY 1

**Pre-emergent flame weeding (when beds are available 7 to 10 days before the seeding date).
Pass through beds with the broadfork.**

- Remove large debris from the bed, if needed (use flail mower, rake or manually remove the large chunks).
- Uniformly apply the alfalfa meal (2-0-2) and pass with the rotary harrow at a depth of 1" (2.5 cm).
- Rake one last time, removing small rocks and debris. The bed must be completely clean.
- Cover the beds with floating row cover to promote the growth of weeds.
- Install sprinklers and keep beds well moistened at all time.
- 7 days later, seed using the Jang 3X. Check seeding density by measuring the weight of seeds distributed.
- Place a handful of beet seeds (as an indicator) at the end of one of the beds and identify with a flag. The beets are the control plants which will indicate when to flame weeds (under the same conditions, beets always germinate before the carrots). Set an alarm or reminder to yourself to check daily, after the 8th day, to see if the beets have emerged. For the seeding to be successful, it is crucial not to miss this indicator.
- As soon as the indicator seeds germinate, flame weed the beds. Adjust the flame weeder to full throttle and move forward slowly and evenly. Your carrots will now have a important head start on any weeds.

ITINERARY 2

Seeding on compost (when beds are not available 7-10 days before seeding).

Pass through beds with the broad fork

- Remove large debris from the bed if needed (use flail mower, rake or manually remove the large chunks).
- Pass the rotary harrow at a 1" (2.5 cm) depth. If there still is debris left, rake again. The goal is to have a very flat and clean surface ready to receive the compost.
- Apply a 1" (2.5 cm) layer of plant-based, weed-free compost on the whole width (30"/75 cm) of the bed. Using the back of the rake, level the compost, making sure that it is the same thickness across the whole width of the bed. Success relies greatly on how carefully this task is accomplished. It's important not to mix the compost with the soil (otherwise weed seeds can emerge).
- Seed using the Jang 3X. Check seeding density by measuring the weight of seeds distributed
- Install sprinklers and keep beds well moistened at all times over the next 10 days.

WEEDING

- If flame-weeded or seeded onto compost, weeds generally won't be able to compete with your crop. We nevertheless cultivated our carrot beds once in order to keep them really clean.
- 20–25 days after direct seeding, when the carrots are at least 3–4" (8-10cm), use bio-disc to cultivate on the row and between the rows in one single pass. Hand pull the bigger weeds.
- After that point, we don't spend time weeding any further. Weeds that emerge after that point shouldn't compete with the crop or have time to go to seed before the carrots are harvested, and are therefore a low priority for weed control.

PLANT PROTECTION

- No action is required

HARVEST

- Carrots are harvested with their tops and bunched in the fields in a kneeling position.
- Using a broadfork, begin by loosening the soil along the rows of carrots.
- One person pulls the carrots, lining them beside each other, while another starts grading and bunching them.
- Generally speaking, we aim to have carrots that are 5" to 7" (13 cm to 18 cm) with approximately 8 carrots per bunch. The size of the carrots affects the number of carrots/bunch (between 8 and 12), but bunches must ALWAYS be the same size.
- We also bunch smaller carrots in which case there are approximately 12 carrots per bunch. All bunches must be the same size.
- For carrots over 7", remove their tops in the fields and collect them in separate bins. They will sell as juicing carrots.
- Don't leave carrots too long in the fields before bringing them in. If the soil dries up on them, they will take longer to clean up afterwards.
- When placing the carrots in the harvest containers, place all bunches in the same direction. Before being conditioned, spray the carrots heavily with the pressure gun (without bruising the leaves since all the leaves are in the same direction).
- 1 person should harvest 60 bunches per hour.
- Once bed is cropped out, cover with tarps in preparation for the next crop.

CONDITIONING

- Carrots are washed using a pressure gun. Upon arriving from the fields, the bins are sprayed and stacked to the left of the sinks.
- The bunches are washed with a pressure gun one by one and piled on the right. Once the pile has reached a certain height, place them in storage bins and identify them.
- Stack identified storage bins on a wheel dolly and bring them to the cold room.
- 1 person should wash and package at least 100 bunches per hour.

STORING

- Fresh carrots are stored in cold room — keep at 35 °F (2 °C)